Name the force and its source

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify forces on a chosen object and name each interaction partner.

SUCCESS CRITERIA - I can...

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart travels west at 12 m/s and west at 4 m/s 3 s later. Is it speeding up? Find acceleration size and direction.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_g = mg$.
F_g is weight; m is mass. Use g = 10 N/kg for CPA. Name the source of every force.

Worked reasoning: A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

Your turn: Draw the book's force diagram. Name the force on Earth paired with the book's weight.

Add and label force arrows.

Book resting on a level table; draw forces on the book only.

Marzano check - circle: ① ② ③

03 Which object gets the force?

A book rests on a level table. Draw its two vertical forces and name who exerts each.

Add and label force arrows.

The dot represents the object. Chosen object: book

Marzano check - circle: ① ② ③

04 Mass and weight

A 2 kg bag hangs at rest from a vertical rope. Use g = 10 N/kg. Draw the forces on the bag, then find its weight and the rope tension. Label each force with its source.

Add and label force arrows.

Chosen object: hanging bag.

Marzano check - circle: ① ② ③

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Continue down each column.

05 After the throw

A ball is moving upward after it leaves your hand. Ignore air resistance. Draw its force diagram and name the source of every force. Explain why upward motion does not require an upward force.

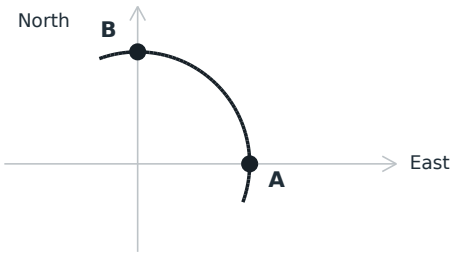
Add and label force arrows.

Ball after release; velocity is upward.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A rover travels straight east at a steady 5 m/s. A second rover turns at the same steady speed. Which can have zero net force? Explain using velocity.
Use the curve for the turning case; show how velocity changes.



Path sketch: runner moves from A toward B.

Marzano check - circle: 1 2 3

07 Exit check

A 3 kg lamp hangs still from one vertical cord. Use g=10 N/kg. Name both forces with source, magnitude and direction.

Add and label force arrows.

Hanging object at rest: label sources and magnitudes.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A hand presses down on the resting book. Add the new force and explain how the table's support must change.

Add and label force arrows.

Book pressed down by a hand: draw forces on the book.

Marzano check - circle: 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.